

# SHIZHE “ANDY” ZHANG

shizhe\_zhang@berkeley.edu · (+1) 510-646-3962 · <https://minikouda.github.io/>  
Berkeley, California, USA

## Education

**University of California, Berkeley** | **GPA: 4.0/4.0** 08.2025 - 12.2026

M.A. in Statistics & Data Science

Advanced topics in Neural Networks, NLP, Randomness & Computation, Causal Inference, Statistical Learning Theory, Optimization

**Zhejiang University** | **GPA: 3.96/4.0** 08.2021 - 06.2025

B.S. in Economics (*Advanced Class*, selected from top 10%)

Game Theory, Incentive Theory, Econometrics, Statistical Software, Time Series, Quantitative Investment, Big Data Analysis

**Minor** (Mathematics): Real Analysis, Stochastic Process, Numerical Analysis, PDE, Operations Research, Data Structures

**Honors:** First Prize in National College Mathematics Competition, Certificate of the Chu Kochen Honors Program (top 1.3%), First Prize Scholarship for Outstanding Economics Students (top 10%), Outstanding Graduate, College-level Scholarship

## Skills

- **Programming:** Python (PyTorch, HuggingFace, scikit-learn), TypeScript, R, SQL (MySQL, PostgreSQL), Matlab, Shell
- **Tools & Software:** Git, Docker, FastAPI, Next.js/React, Cloudflare Workers, Cloud (AWS, GCP), Unix, LaTeX
- **Technical Skills:** Machine Learning, Deep Learning, Retrieval & Ranking, Feature Engineering, Causal Inference, Model Evaluation, Optimization, NLP, LLM/VLM Agents, RAG, MCP, vLLM, Structured Decoding, LLM-as-Judge

## Internships

**Monolithic Power Systems, AI Developer** 05.2026 - 08.2026

- Built a layout-preserving translation agent for PDF/DOCX/PPTX. Designed a multi-level memory: a short-term window for local coherence, retrieved exemplars from past translations, and proper-noun records pinning terms document-wide. Per-language-pair LoRA adapters hot-swap on self-hosted vLLM, raising COMET  $\sim 3$  points across all pairs.
- Designed an in-house SEC EDGAR MCP server for the finance team with edge caching and grounded, cited figures.
- Engineered a slide-generation agent compiling typed actions (charts, tables, Mermaid) via python-pptx. One action registry generates the model's schema, runtime validation, and execution dispatch. Every edit appends to an immutable log, giving decks undo and branching version history; content quality rose 20% on a PPTEval-style rubric.

**SDIC Securities, Quantitative Researcher** 10.2023 - 03.2024

- Developed event-driven strategies for A-share stocks on the Billboard (*Dragon-Tiger List*), framing short-term price movement as a supervised prediction problem.
- Engineered features from raw trade data, including net inflow strength, institutional concentration, and order imbalance.
- Trained a LightGBM model to rank stocks by predicted short-term return; the top-quintile portfolio earned a 2.3% monthly excess return over the market benchmark.
- Built a capital-activity factor by modeling co-trading behavior across brokerages with network analysis; in a long-short strategy (CSI 300 vs. CSI 2000), it improved signal precision by 14% over heuristic benchmarks.

## Projects

**Cost-Aware Clarification in Multimodal Reference Games** 04.2026 - 05.2026

- Derived a training-free expected-utility rule for when a vision-language listener should ask versus commit (commit iff top posterior  $\geq 1 - c$ ); built the speaker/listener framework (Claude, Gemini VLM APIs), evaluation pipeline, and metrics suite.
- Ran 28,800 zero-shot trials across 8 speaker and 6 listener strategies over 600 rendered scenes, scoring cost-penalized accuracy against clarification rate.
- Showed the clarification rule lowers cost-penalized accuracy for every probabilistic listener (removing it gained +0.59 for CoT); posterior sharpening recovered 76% of that gap, identifying uncertainty calibration as the key lever.

**RAG-Driven Company Search System (Industry Partner: Pyramyd)** 10.2025 - 12.2025

- Built a RAG-based discovery system over 1M+ reviews with hybrid candidate fusion and metadata reranking, raising retrieval relevance by 30–50% with real-time ( $< 1s$ ) retrieval and 40% shorter LLM prompts.
- Fine-tuned a bi-encoder retriever with contrastive (InfoNCE) loss, lifting NDCG@10 by  $\sim 70\%$  on held-out companies.
- Benchmarked exact FAISS, HNSW and IVF-PQ indexes with a cross-encoder reranker: HNSW kept 96% of exact-search quality at  $< 1$  ms p95, and reranking the top 50 raised NDCG@10 by  $\sim 17\%$  over exact dense search at 64 ms p95.
- Validated robustness with counterfactual ablations across 10+ pipeline variants: top- $k$  rankings stayed 85–90% stable, and removing key components cut recall by 20–35%.

**Robust Portfolio Optimization of CNE5 Factors** 11.2024 - 12.2024

- Standardized 100+ raw style/industry factors, applied sparse PCA factor selection to derive interpretable and stable factor structures, and estimated factor covariance matrices under high dimensionality.
- Optimized portfolios using mean-variance and CVaR minimization under sector-neutrality and turnover constraints.
- Evaluated robustness via 10,000+ Monte Carlo stress scenarios and block bootstrap resampling.
- Benchmarked Barra-style linear risk models against ML-enhanced covariance estimation, including Ledoit-Wolf shrinkage and Graphical Lasso for high-dimensional stability, improving Sharpe ratio by 11% and reducing max drawdown by 6%.